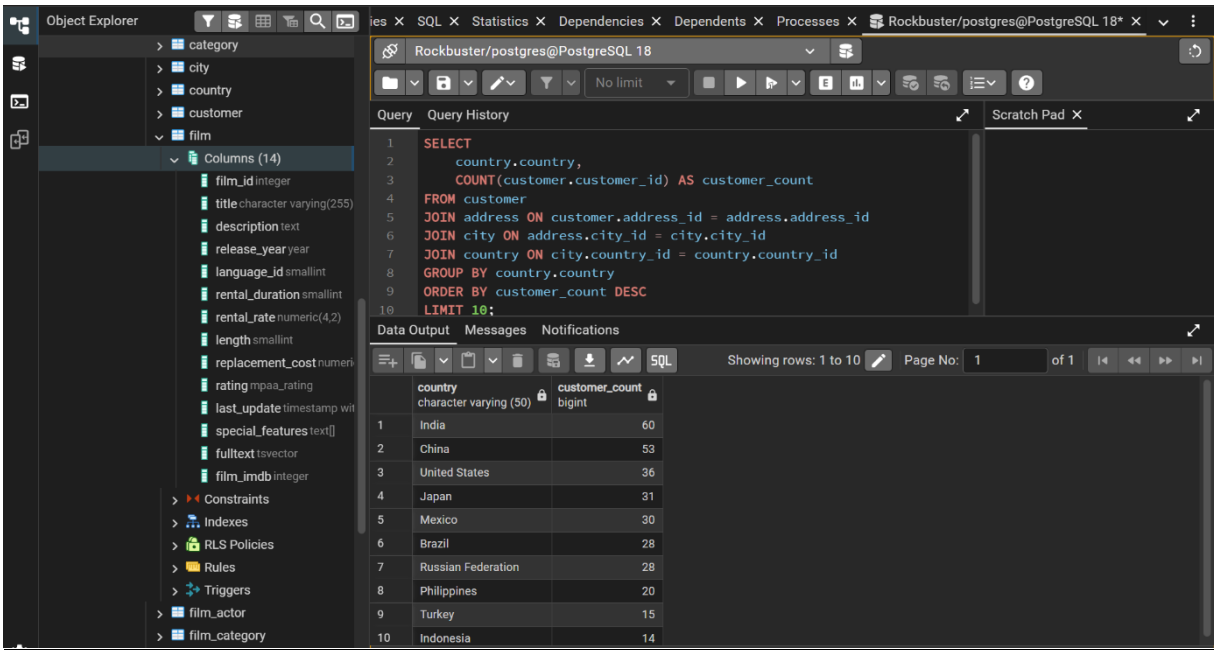
# Joining Tables of Data

Write a query to find the top 10 countries for Rockbuster in terms of customer numbers.




**Query Applied:**

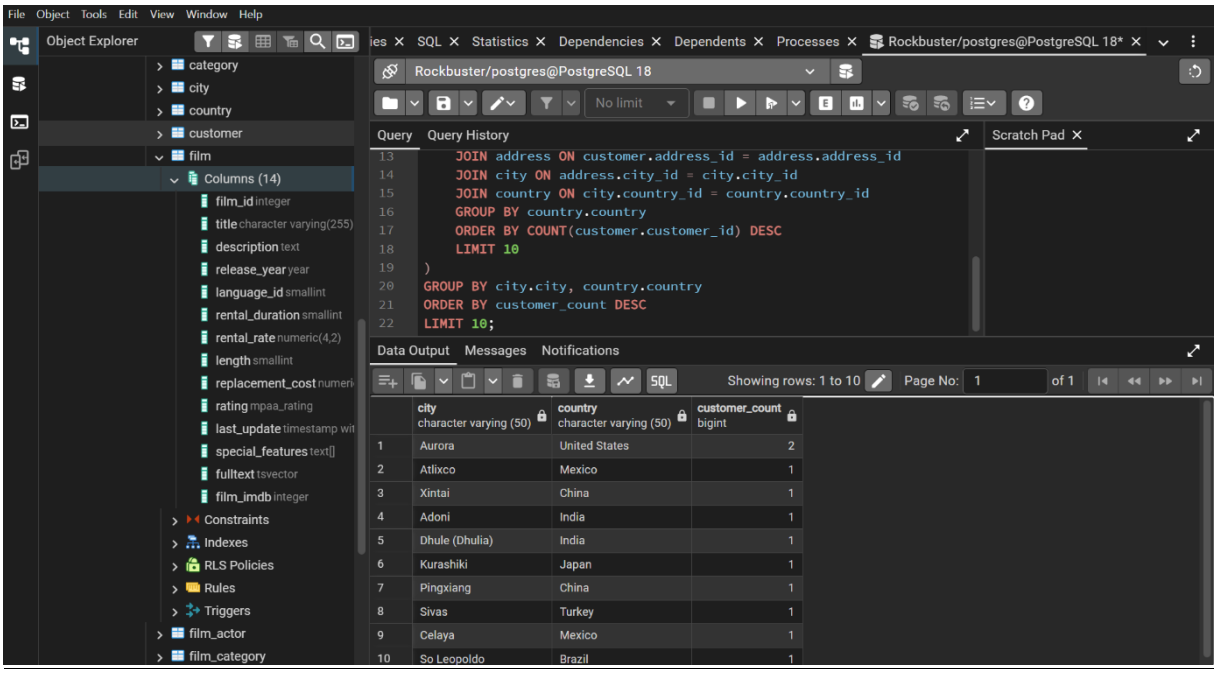
```
SELECT
   country.country,
   COUNT(customer.customer_id) AS customer_count
FROM customer
JOIN address ON customer.address_id = address.address_id
JOIN city ON address.city_id = city.city_id
JOIN country ON city.country_id = country.country_id
GROUP BY country.country
ORDER BY customer_count DESC
LIMIT 10;
```

Write a few sentences on how you approached this query and why. You must be able to explain your thought process when writing queries, especially for future interviews.

I first identified which tables contain country information. Since the customer table does not store country directly, I joined the related tables following the logical relationship: customer → address → city → country

After establishing the joins, I grouped the results by country and used COUNT(customer.customer_id) to measure how many customers belong to each country. I then sorted the results in descending order using ORDER BY and restricted the output to the top 10 using LIMIT. This approach ensures the aggregation is performed correctly at the country level.

Next, write a query to identify the top 10 cities that fall within the top 10 countries you identified in step 1. (Hint: the top 10 cities can be in any of the countries identified—you don't need to create a separate list for each country.)




**Query Applied:**

```
SELECT
    city.city,
    country.country,
    COUNT(customer.customer_id) AS customer_count
```

```
FROM customer
JOIN address ON customer.address_id = address.address_id
JOIN city ON address.city_id = city.city_id
JOIN country ON city.country_id = country.country_id
WHERE country.country IN (
  SELECT
    country.country
  FROM customer
  JOIN address ON customer.address_id = address.address_id
  JOIN city ON address.city_id = city.city_id
  JOIN country ON city.country_id = country.country_id
  GROUP BY country.country
  ORDER BY COUNT(customer.customer_id) DESC
  LIMIT 10
)
GROUP BY city.city, country.country
ORDER BY customer_count DESC
LIMIT 10;
```

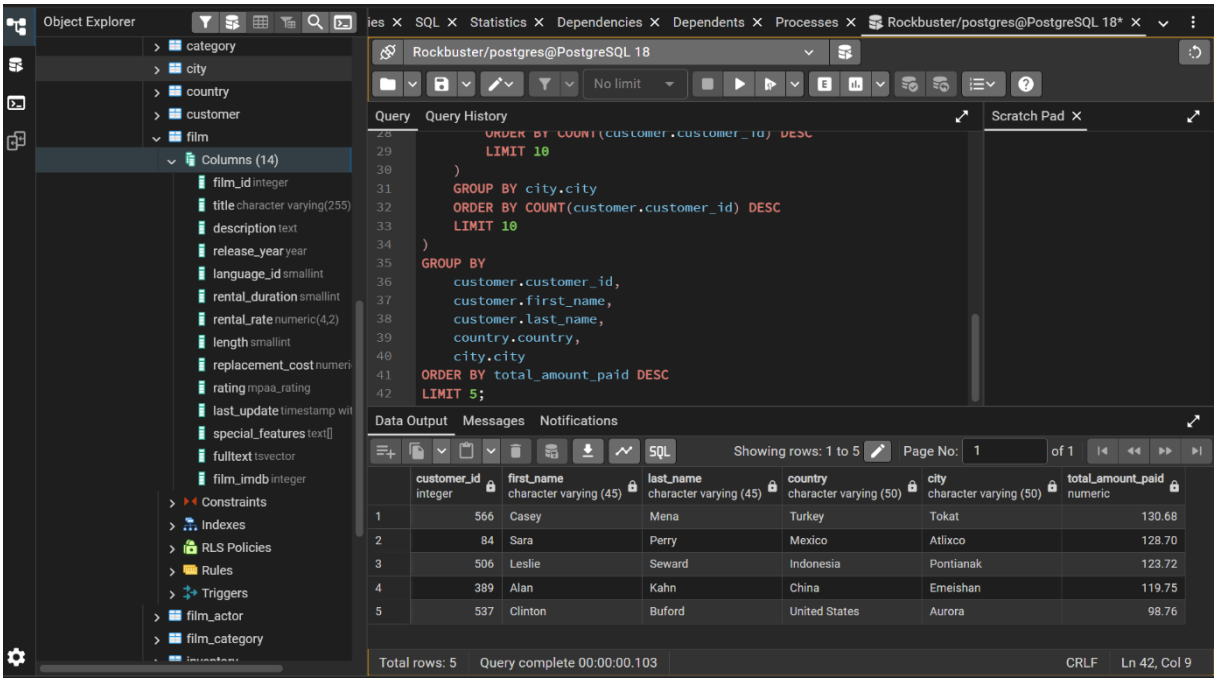
Write a short explanation of how you approached this query and why.

For this query, I reused the same join structure because city and country information are obtained through the same relationships.

To restrict results to only the top 10 countries, I used a subquery inside the WHERE clause. This subquery dynamically returns the highest-ranking countries based on customer count. After filtering by country, I grouped the data by city and counted customers per city. Finally, I sorted the results and selected the top 10 cities.

Using a subquery keeps the logic flexible and avoids hard-coding country names.

Now write a query to find the top 5 customers from the top 10 cities who've paid the highest total amounts to Rockbuster. The customer team would like to reward them for their loyalty!




**Query Appled:**

```
SELECT
    customer.customer_id,
    customer.first_name,
    customer.last_name,
    country.country,
    city.city,
    SUM(payment.amount) AS total_amount_paid
FROM customer
JOIN payment ON customer.customer_id = payment.customer_id
JOIN address ON customer.address_id = address.address_id
JOIN city ON address.city_id = city.city_id
JOIN country ON city.country_id = country.country_id
WHERE city.city IN (
```

```
  SELECT
    city.city
  FROM customer
  JOIN address ON customer.address_id = address.address_id
  JOIN city ON address.city_id = city.city_id
  JOIN country ON city.country_id = country.country_id
  WHERE country.country IN (
    SELECT
      country.country
    FROM customer
    JOIN address ON customer.address_id = address.address_id
    JOIN city ON address.city_id = city.city_id
    JOIN country ON city.country_id = country.country_id
    GROUP BY country.country
    ORDER BY COUNT(customer.customer_id) DESC
    LIMIT 10
  )
  GROUP BY city.city
  ORDER BY COUNT(customer.customer_id) DESC
  LIMIT 10
)
GROUP BY
  customer.customer_id,
  customer.first_name,
  customer.last_name,
  country.country,
  city.city
ORDER BY total_amount_paid DESC
LIMIT 5;
```

**Thoughts:**

This query builds on the previous logic but adds customer spending behavior via the payment table. I joined the payment table to calculate total revenue per customer using SUM(payment.amount). To ensure only customers from the top cities are considered, I used nested subqueries that first identify:

1. Top 10 countries
2. Top 10 cities within those countries

After filtering the customers, I grouped by customer details and sorted by the total amount paid in descending order. Finally, I limited the results to the top 5 customers.

This structured filtering approach ensures accuracy while keeping the query scalable.